

39 The average sizes of some pathogens are shown.

pathogen	average size /nm
HIV particle	100
<i>Mycobacterium tuberculosis</i>	3 000
<i>Plasmodium</i>	20 000
<i>Vibrio cholerae</i>	1 500

One type of air filter has been shown to be effective at preventing any pathogens of 1 μm or larger from entering the air system of a room.

Based on their size **and** mode of transmission, which diseases would the air filter prevent from entering the air system of a room?

	HIV infection	TB	malaria	cholera
A	x	✓	✓	✓
B	x	x	✓	✓
C	✓	x	x	x
D	x	✓	x	x

key

✓ = prevented

x = not prevented